NET2008 DevOps: Practical Assessment 4

Pipeline Incident Audit | Week 15, in class, 2 hours | individual | 100 marks

The situation

You join a team that owns a small Python service. It has tests, a Dockerfile, a compose file, a GitHub Actions workflow, a Jenkinsfile, a Nagios file and an Ansible playbook. Nothing works as it should. Your job is to run an **incident audit**: find every fault, fix it, and explain it in writing. This is the same kind of work you did in Weeks 9 to 14, now on someone else's files, under time pressure.

What you receive

At the start of class your instructor gives you a kit named `pa4-kit-seed-NN.zip`. `NN` is **your seed**: the sum of the last four digits of your student ID, written with two digits (for example 17 or 09). Use only your own kit. A kit for a different seed is marked as failed.

What you do

- Unzip the kit, enter the folder `pa4-kit`, and run `git init -b main`. Commit the files as they are: that is your starting point.
- Eight faults are hidden in the kit. They can be in the Python code, the workflow, the Jenkinsfile, the Dockerfile, the compose file, the Nagios file and the playbook. You are **not told how many are in which file**.
- Fix each fault with its **own commit**. The commit message names the file and the problem.
- Do **not** edit anything in `tests/`. The tests are correct. A change to a test is marked as a failed fix.
- Do not delete a file or a whole section to make a problem go away. Keep the structure.
- Write `INCIDENT.md` with **one section per fault**, in this form:

```
## Fault 1: short title
Symptom: what you saw (error message, wrong output, or what you noticed when reading)
Cause:   why it happens, in your own words
Fix:     what you changed, and in which file
Prevent: one thing the team could do so it does not return
```

- Push to a **public** GitHub repository named `pa4-incident-audit` and submit the URL on Brightspace **before the end of class**. Do not push after.

Rules

- Work **alone**, at your own laptop, in a real terminal. The documentation of Docker, Compose, GitHub Actions, Jenkins, Nagios and Ansible, your own repositories from Weeks 9 to 14, and the course repository are allowed.
- No AI tools, no chat, no messages. Your instructor may ask you to close any tab.
- Nothing here needs a running Docker, Kubernetes or Jenkins server. The checker reads your files and runs your tests.
- If you miss the session without a valid reason, the grade is 0.

Marks

Part	What	Marks
Faults fixed	5 marks for each of the 8 faults. The fix must remove the problem and keep the file working.	40
Kit health	Tests unchanged and passing, all original files still present, INCIDENT.md has 8 sections.	10
INCIDENT.md	Read by your instructor: causes are correct and in your own words, prevention is realistic.	20
Live explanation	Right after the session, your instructor asks you about three of your fixes and may ask you to change one thing. You explain without notes and without help.	30

How to work

Run the tests first. Then read the files one at a time and compare each with the version you built in the labs: ask *what would I have written here?* Typical mistakes are small: a wrong name, a missing line, a wrong number, a value in the wrong form. Fix one fault, commit, and move to the next. Leave the last 15 minutes to write INCIDENT.md carefully. A fault you fixed but cannot explain is worth little in the live explanation.